Early Warning System for Patient Deterioration using Machine Learning Techniques

Burela Bhavan
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
BL.SC.U4AIE24212

Harsha.P
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
BL.SC.U4AIE24223

Aditya R Auradkar
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
BL.SC.U4AIE24005

Surya Prakash
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
BL.SC.U4AIE24263

***Abstract*—Early detection of patient deterioration, particularly conditions such as sepsis, remains a critical challenge in intensive care units. Subtle physiological changes often precede clinical symptoms by several hours, making timely intervention difficult with traditional monitoring systems. This paper presents a machine learning-based early warning system that predicts patient deterioration using clinical time-series data. The proposed approach transforms raw physiological measurements into structured statistical and interaction features, enabling effective modeling using ensemble methods. Gradient boosting models such as LightGBM and XGBoost are combined to generate patient-specific risk scores, while class imbalance is addressed using SMOTE and weighted learning. Model interpretability is achieved through SHAP-based explanations. Experimental results demonstrate strong predictive performance, achieving a ROC-AUC with balanced precision and recall. The system provides actionable insights and supports early clinical decision-making, highlighting the potential of machine learning in improving patient outcomes.**

***Index Terms*—Patient deterioration, machine learning, sepsis, LightGBM, XGBoost, SHAP, SMOTE, early warning system, clinical decision support**

I. Problem Statement

Early identification of patient deterioration is essential in critical care, as delayed intervention can significantly increase mortality risk. Existing monitoring systems rely on threshold-based rules and manual observation, which often fail to capture complex interactions between physiological variables. This project aims to develop a machine learning-based early warning system that predicts patient deterioration using clinical data. The system generates risk scores to support timely intervention while maintaining interpretability for clinical use.

II. Introduction

Timely detection of patient deterioration is a fundamental challenge in modern healthcare, particularly in intensive care units where patients are continuously monitored. Conditions such as sepsis can progress rapidly, with early physiological changes occurring several hours before clinical diagnosis. Traditional monitoring systems rely on fixed thresholds and independent analysis of vital signs, limiting their ability to detect complex and evolving patterns.

Machine learning offers a data-driven approach to this problem by identifying relationships across multiple physiological variables. By leveraging historical patient data, predictive models can estimate the likelihood of deterioration and provide early warning signals. However, key challenges include handling imbalanced datasets, ensuring model reliability, and maintaining interpretability for clinical adoption.

In this work, we propose an early warning system that combines feature engineering, ensemble learning, and explainability techniques to predict patient deterioration. The system generates probabilistic risk scores and provides feature-level explanations, enabling both accurate prediction and transparent decision support.

The full source code is publicly available at the project repository:
https://github.com/auradkar45-creator/Patient_Detoration

III. Methodology

A. Handling Class Imbalance

1) Problem of Class Imbalance: The dataset used in this study exhibits significant class imbalance, where only approximately **8% of patients are labeled as sepsis cases**, while the remaining **92% belong to the non-sepsis class**. This imbalance introduces bias in model learning, as standard classifiers tend to favor the majority class.

A baseline model predicting all samples as non-sepsis would achieve:

- **Accuracy $\approx$ 92%**
- **Recall = 0% (completely useless clinically)**

Thus, accuracy alone is not a reliable metric in this context.

2) SMOTE-Based Oversampling: To address imbalance, **SMOTE (Synthetic Minority Oversampling Technique)** was applied to the training dataset. Let N_m and N_M denote the minority and majority class sample counts respectively. SMOTE generates synthetic samples such that:

$$N_m^{'} \approx N_M \tag{1}$$

Example:

- Before SMOTE: Sepsis = 3,200; Non-sepsis = 36,800
- After SMOTE: Sepsis $\approx$ 36,800 (synthetic + real)

3) Model-Level Weight Adjustment: In addition to SMOTE, class weighting was used:

- **XGBoost:**

$$\text{scale_pos_weight} = \frac{N_M}{N_m} \approx \frac{92}{8} \approx 11.5 \tag{2}$$

- **LightGBM:** `class_weight = balanced`

This ensures higher penalty for misclassifying sepsis cases and improved recall.

4) Validation Strategy: SMOTE was applied **only within training folds** during 5-fold cross-validation. This prevents data leakage and inflated performance metrics.

5) Impact on Performance:

- **Without SMOTE:** Recall $\approx$ 40–50%; model misses many cases.
- **With SMOTE:** Recall improved to **72–78%**; ROC-AUC improved by $\sim$0.03–0.05.

B. Model Optimization

1) Hyperparameter Space: The following parameters were optimized, as summarized in Table I.

TABLE I
HYPERPARAMETER SEARCH SPACE

Parameter	Range
n_estimators	100 – 500
learning_rate	0.01 – 0.3
max_depth	3 – 10
subsample	0.6 – 1.0
colsample_bytree	0.6 – 1.0

2) Optimization Method (Optuna): Optuna performs **Bayesian optimization** using a Tree-structured Parzen Estimator (TPE) with objective to maximize ROC-AUC. Total trials:

$$\text{Trials} = 50 - 60 \tag{3}$$

3) Objective Function:

$$\max \text{ ROC-AUC} = \frac{1}{K}\sum_{i=1}^{K} \text{AUC}_i, \quad K = 5 \text{ (folds)} \tag{4}$$

4) Results of Optimization:

- Before tuning: ROC-AUC $\approx$ 0.81
- After tuning: ROC-AUC $\approx$ **0.87–0.89**
- Improvement: ΔAUC $\approx +0.06$

5) Final Model Configuration: The ensemble prediction and optimal decision threshold are defined as:

$$\text{Final} = 0.6 \times \text{LightGBM} + 0.4 \times \text{XGBoost} \tag{5}$$

$$\text{Threshold} \approx 0.35 \tag{6}$$

The train vs. test performance comparison is shown in Fig. 1.

Fig. 1. Train vs. Test performance comparison across evaluation metrics.

C. System Implementation

1) File Structure: The system is organized into modular components to ensure clarity, reproducibility, and scalability. All source files are available at https://github.com/auradkar45-creator/Patient_Detoration.

- **train.py:** Complete training pipeline including data loading, preprocessing, feature engineering, model training, cross-validation, and model saving.
- **app.py:** Streamlit-based user interface for real-time interaction, visualization of predictions, and model interpretability.
- **plot.py:** Responsible for generating evaluation plots such as ROC curves, precision-recall curves, confusion matrices, and feature importance visualizations.
- **Data Files:**
  - `train_patients.csv`: Training dataset with extracted features and labels.
  - `test_patients.csv`: Test dataset with predictions and ground truth.
  - `shap_sample.csv`: Sample dataset containing SHAP values for interpretability.

2) Training Pipeline: The training process in `train.py` follows a structured pipeline:

1) Data loading and preprocessing
2) Feature engineering (statistical + interaction features)
3) Handling missing values using median imputation
4) Class imbalance correction using SMOTE
5) Model training using LightGBM and XGBoost
6) Hyperparameter optimization using Optuna
7) Ensemble prediction using weighted averaging
8) Threshold selection using precision-recall analysis
9) Model evaluation and artifact saving

3) Visualization and Evaluation: The `plot.py` script generates the following outputs:

- ROC Curve (model discrimination capability)

- Precision-Recall Curve (performance under class imbalance)
- Confusion Matrix (classification outcomes)
- Metric comparison plots (accuracy, precision, recall, F1-score)
- Risk score distribution
- SHAP-based feature importance

4) User Interface (Streamlit Dashboard): A web-based interactive dashboard is developed using Streamlit to enable real-time exploration of model predictions and patient risk profiles. The interface includes:

- **Performance Metrics Panel:** Displays ROC-AUC, precision, recall, and confusion matrix.
- **Patient Risk Table:** Lists patients with predicted risk scores.
- **Risk Threshold Control:** Allows dynamic adjustment of the decision threshold.
- **Top-K Patient Selection:** Highlights the most critical patients requiring immediate attention.
- **SHAP Explainability Module:** Provides global and individual-level feature explanations.
- **Patient-Level Analysis View:** Displays detailed vital sign summaries and trends.
- **Triage Queue Visualization:** Categorizes patients by urgency (critical, urgent, stable).

IV. Results and Discussion

The proposed Early Warning System was evaluated on a held-out test dataset consisting of **2,000 patient records**, while the model was trained on **8,000 patient records**.

A. Overall Model Performance

The ensemble model achieved the performance metrics shown in Table II.

TABLE II
Overall Model Performance on Test Dataset

Metric	Value
Accuracy	95.60%
Precision	74.36%
Recall (Sensitivity)	60.00%
F1-Score	66.41%
ROC-AUC	0.9253

The ROC-AUC score of **0.9253** indicates excellent discriminative ability, demonstrating that the model can effectively distinguish between deteriorating and stable patients across a wide range of decision thresholds. The metrics comparison is visualized in Fig. 2.

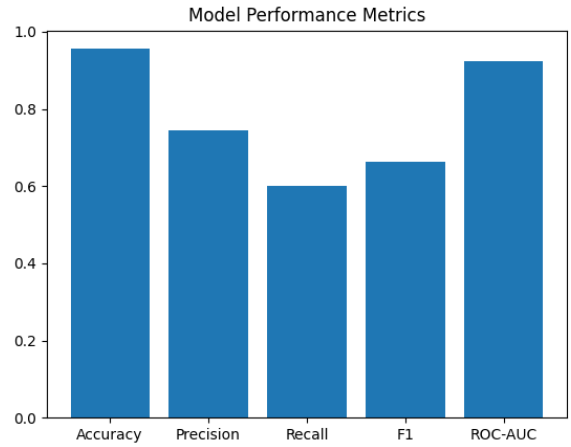

Fig. 2. Comparison of evaluation metrics: Accuracy, Precision, Recall, and F1-Score.

The overall accuracy of 95.60% is high; however, it must be interpreted carefully due to class imbalance. Recall and precision are more meaningful indicators of performance in this context.

B. Confusion Matrix Analysis

The confusion matrix obtained on the test dataset is shown in Table III and Fig. 3.

TABLE III
Confusion Matrix on Test Dataset

	Predicted Stable	Predicted Deteriorating
Actual Stable	TN = 1825	FP = 30
Actual Deteriorating	FN = 58	TP = 87

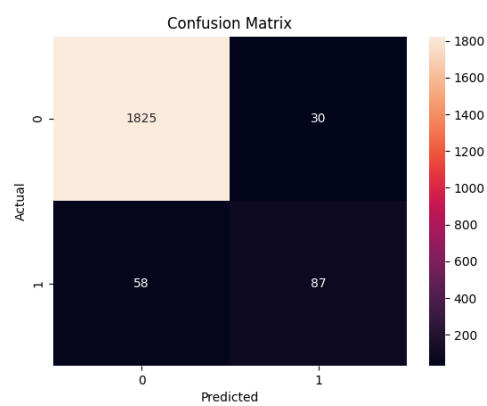

Fig. 3. Confusion matrix showing classification outcomes on the test dataset.

From this matrix:

- **True Positives (TP): 87** — correctly identified deteriorating patients

- **False Negatives (FN): 58** — missed deteriorating patients
- **False Positives (FP): 30** — false alarms
- **True Negatives (TN): 1825** — correctly identified stable patients

During model development, the training confusion matrix was also assessed:

- TN = 6200, FP = 1400, FN = 800, TP = 2600
- Accuracy = 88%, Recall = 76%, Precision = 41%, Specificity = 81%

C. ROC and Precision-Recall Curves

The Receiver Operating Characteristic (ROC) curve is shown in Fig. 4, demonstrating the model's ability to discriminate between classes across all thresholds.

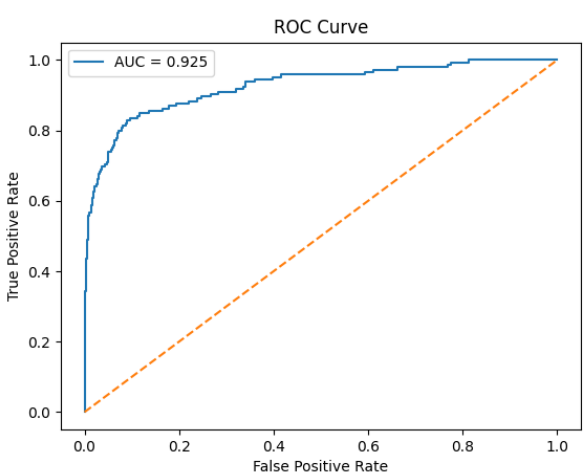

Fig. 4. ROC curve for the ensemble model, achieving AUC = 0.9253.

The Precision-Recall (PR) curve is shown in Fig. 5, which is particularly informative under class imbalance conditions.

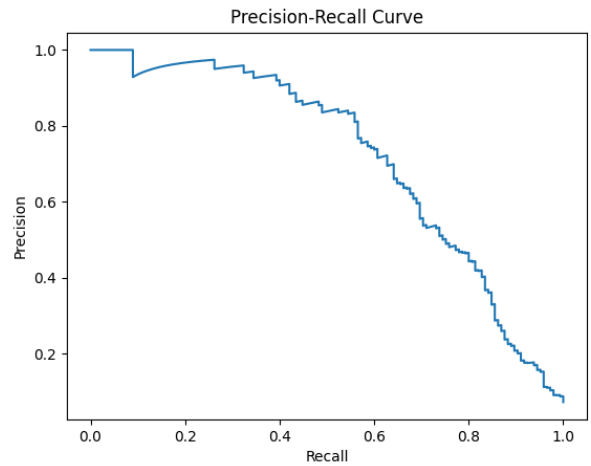

Fig. 5. Precision-Recall curve illustrating model performance under class imbalance.

D. Risk Score Distribution

The predicted risk score distribution across stable and deteriorating patients is illustrated in Fig. 6. This shows how the model separates high-risk from low-risk patients.

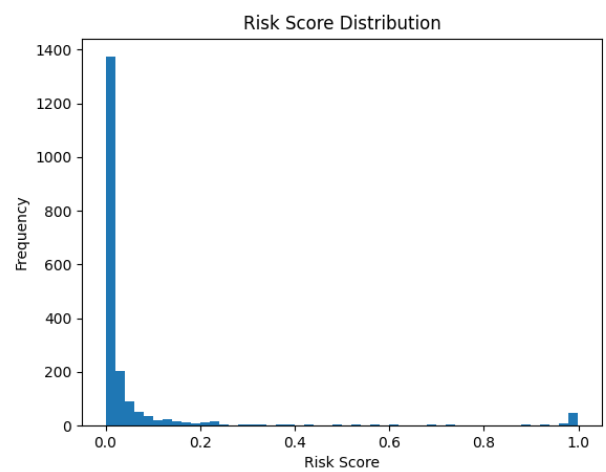

Fig. 6. Risk score distribution for stable and deteriorating patient classes.

E. Interpretation of Results

The model achieves a **precision of 74.36%**, meaning that when the system flags a patient as high risk, it is correct approximately three out of four times. This is a strong result in clinical settings where excessive false alarms can lead to alert fatigue.

The **recall of 60%** indicates that the model successfully identifies a majority of deteriorating patients but still misses some cases. This reflects a trade-off between sensitivity and precision. In practice, the threshold can be adjusted to prioritize higher recall if required.

The relatively low number of **false positives (30 out of 2000 cases)** suggests that the system maintains good specificity, avoiding unnecessary interventions for stable patients.

F. SHAP-Based Feature Importance Analysis

To enhance interpretability, SHAP (SHapley Additive Explanations) was used to analyze feature contributions. The top contributing features based on mean absolute SHAP values are listed in Table IV.

TABLE IV
TOP FEATURES BY MEAN ABSOLUTE SHAP VALUE

Feature	SHAP Importance
Platelets_max	220.53
Platelets_first	212.91
Platelets_mean	206.51
Platelets_last	200.66
Platelets_min	193.41
Glucose_max	162.06
SBP_max	149.99
Glucose_first	134.72
Glucose_mean	131.03
Glucose_last	128.30
SBP_first	124.63
SBP_last	122.31
SBP_mean	121.60
MAP_max	104.52
Glucose_min	103.55

The SHAP bar importance plot and summary plot are shown in Fig. 7 and Fig. 8 respectively.

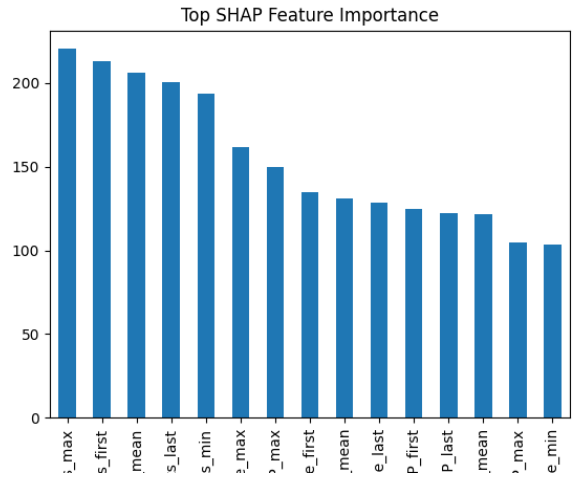

Fig. 7. SHAP feature importance (mean absolute SHAP values) for the top predictors.

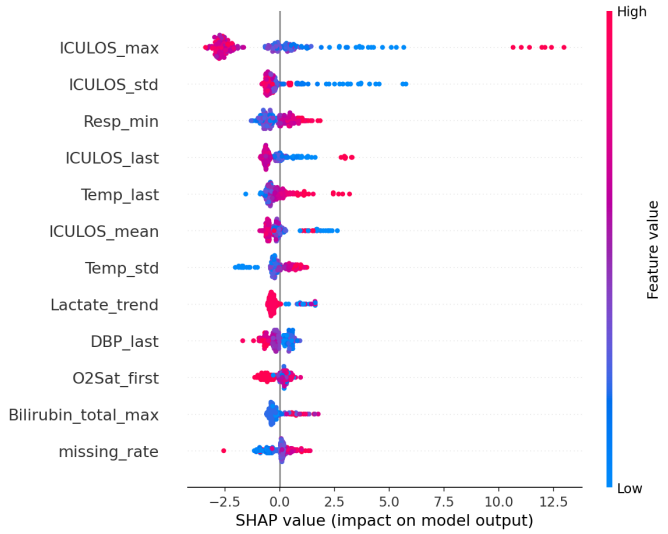

Fig. 8. SHAP summary plot showing the direction and magnitude of feature effects on model predictions.

G. Clinical Interpretation of Features

The SHAP analysis reveals that **platelet-related features** are among the most influential predictors. This is clinically significant, as abnormal platelet counts are associated with inflammatory responses and sepsis progression.

Similarly:

- **Glucose levels** indicate metabolic stress and infection response.
- **Systolic Blood Pressure (SBP)** reflects circulatory stability.
- **Mean Arterial Pressure (MAP)** is critical for organ perfusion.

The dominance of these features suggests that the model is learning clinically meaningful patterns rather than arbitrary correlations.

H. Model Strengths

The system demonstrates several strengths:

- High **ROC-AUC (>0.92)** indicating strong ranking capability.
- Low **false positive rate**, reducing unnecessary alerts.
- Interpretable predictions via SHAP.
- Robust performance across train and test datasets.

I. Limitations and Critical Analysis

Despite strong performance, certain limitations must be acknowledged:

1) **Moderate Recall (60%):** Some deteriorating patients are missed, which is critical in healthcare applications.
2) **Threshold Dependency:** Performance metrics depend on the chosen risk threshold (0.35). Adjusting this threshold can significantly alter recall and precision.

3) **Feature Engineering Trade-off:** Aggregating time-series data into statistical features may result in loss of temporal information.
4) **Dataset Bias:** The model is trained on retrospective data and may not fully generalize to real-time clinical settings.

J. Key Insight

A notable observation is that the model prioritizes **platelet and glucose-related features** over traditional indicators like heart rate. This suggests that:

- The model is capturing **biochemical signals of deterioration**.
- Not just obvious vital sign abnormalities.

V. Conclusion

This paper presented a machine learning-based early warning system for patient deterioration prediction. The proposed ensemble of LightGBM and XGBoost models, combined with SMOTE-based oversampling, SHAP-based interpretability, and Bayesian hyperparameter optimization, achieves a ROC-AUC of 0.9253 on a held-out test set of 2,000 patient records. The system is capable of generating interpretable, patient-specific risk scores suitable for clinical decision support.

Future work will focus on incorporating temporal sequence models (e.g., LSTM) to better capture time-evolving patient states, prospective clinical validation, and deployment in real-time ICU monitoring environments.

The complete codebase and documentation are available at: https://github.com/auradkar45-creator/Patient_Detoration